

Thomas Zhang

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Software Engineering

Sep. 2024 - Apr. 2028

- Rover Autonomy Developer at Watonomous design team, Percussionist at Orchestra@UWaterloo
- Relevant Coursework: Data Abstraction & Implementation, Linear Algebra, Calculus I/II

TECHNICAL SKILLS

Languages: C/C++, Python, Java, C#, SQL, JavaScript/TypeScript, HTML/CSS

Frameworks: FastAPI, Flask, PyTorch, OpenCV, NumPy, React.js, React Native, Next.js, Node.js

Tools: Git, GitLab CI/CD, GitHub Actions, Docker, Celery, Redis, Apache Kafka, Pinecone, Google ADK, Gemini API, Supabase, TimescaleDB, PostgreSQL, SQLite

EXPERIENCE

Software Developer Intern

Jun. 2025 – Aug. 2025

Triple J Canada Consulting Inc.

Mississauga, ON

- Built a production full-stack tax platform for **2,000+** clients and **5,000+** returns, replacing paper intake and in-person office visits with web-based data collection
- Delivered **Flask** backend that reduced staff case prep work by **>30%** by shifting data entry to clients
- Streamlined onboarding with client email verification and admin RBAC, eliminating **20+ hrs/week** of work
- Ran live **SQLite** database migrations using Alembic, with **zero loss or downtime**

Software Developer

Jan. 2026 – Present

Watonomous Design Team

Waterloo, ON

- Reduced obstacle costmap input from 307K to ~15K pts/frame via **20× point cloud** downsampling
- Added line-of-sight smoothing to **A*** planner on 60×60 grids, cutting redundant waypoints
- Integrated **YOLOv8** via **ONNX Runtime** on simulated depth camera feeds for obstacle detection

PROJECTS

TradeStream | *C++, Python, FastAPI, Next.js, Apache Kafka, Redis, TimescaleDB, Docker*

- Architected a real-time market data platform processing **1M+ daily ticks** across 100 symbols, streaming live VWAP, volatility, and volume-spike analytics to a **Next.js** dashboard via **WebSockets**
- Ran a **C++** microservice computing rolling analytics over a 20-tick sliding window via librdkafka
- Decoupled tick ingestion, analytics, and WebSocket broadcast across three independent **Kafka** consumer groups
- Routed live ticks to **Redis** for recent access and **TimescaleDB** hypertables for historical retention

JobFlow | *Next.js, FastAPI, Celery, Redis, Scrapy, Playwright, PostgreSQL, Supabase*

- Engineered a scalable job search and tracking platform with **FastAPI** serving **150+ concurrent users**, leveraging asynchronous processing and real-time updates to centralize career listings
- Implemented web-scraping pipeline with 70%+ hit rate using **Scrapy** with **Playwright** and rotating proxies
- Designed distributed task queue with **Celery** workers, **Redis** pub/sub, and WebSocket broadcasting
- Secured API routes with **JWT** authentication and deployed the full stack via **Docker Compose** on Railway

Yapdraw - GenAI Genesis 2026 Finalist | *Next.js, Excalidraw, Deepgram, OpenAI API, IndexedDB*

- Built a AI voice tool that converts spoken system architecture into editable Excalidraw diagrams in real time
- Reduced perceived LLM response time to **< 1s** by firing one **OpenAI API** call per spoken sentence via **Deepgram** WebSocket, so diagrams grow live as the user speaks
- Built a safety layer that detects and restores nodes the LLM accidentally drops between voice edits, keeping diagrams stable across long sessions
- Persisted diagrams to **IndexedDB** with auto-save, diagram version history, and rollback to any prior state

PlannerAI | *Python, Google ADK, Pinecone, Gemini API, Pydantic, PyMuPDF*

- Designed an AI study planner with 6-agent **RAG** architecture using **Google ADK** and **Gemini API** to generate personalized study plans and guides from course textbooks and syllabi
- Implemented generator-critic validation loop, achieving **75%+** retrieval accuracy across curriculum topics
- Utilized parent-child chunking for semantic search via **Pinecone**, optimizing retrieval precision and context
- Optimized performance with embeddings cache and exponential backoff, reducing API calls by **>60%**